

ContextTrace: Evidence-Chain Forensics for Retrieval-Augmented Generation

Anonymous ACL submission

Abstract

Retrieval-augmented generation (RAG) evaluators often reduce a trace to scalar quality scores, leaving developers to reconstruct which claim failed, which evidence mattered, and which component should change. We introduce CONTEXTTRACE, a local-first evidence-chain forensics system that maps answer claims to retrieved spans, citation edges, source conditions, abstention decisions, failure labels, root causes, and repair suggestions. A frozen pre-review evaluation covers 150 public-document diagnostic cases, 200 RAGTruth cases, a same-ID RAGAS baseline, and eleven registered ablation rows. The current system obtains 0.955 failure-label macro-F1 and zero dangerous false greens on the RAGTruth sample, but claim-verdict macro-F1 is 0.337 and independent label review remains incomplete. We additionally run a controlled 1,410-judgment LLM-simulated protocol stress test. The simulated evidence-chain condition improves root-cause accuracy but not mean actionability, so it does not support the preregistered positive RQ4 claim and does not replace human validation. We release a deterministic experiment and anonymous-artifact workflow with fail-closed claim gates.

1 Introduction

Retrieval-augmented generation (RAG) conditions generation on external documents, improving access to knowledge without encoding every fact in model parameters (Lewis et al., 2020). The same decomposition creates a debugging problem. A wrong answer may follow from retrieval failure, evidence that is present but misranked, a citation attached to the wrong source, an unsupported answer extension, conflicting or stale sources, or a failure to abstain. A single faithfulness score identifies none of these repair targets.

We study *evidence-chain forensics*: reconstructing a verifiable path from each answer claim through retrieved evidence and citation edges to an observable failure and a concrete repair. This framing requires more than detecting hallucination. It requires preserving intermediate decisions, refusing to impute diagnostics a baseline does not expose, and distinguishing grounding in supplied evidence from real-world truth.

CONTEXTTRACE implements this framing as a deterministic, local-first pipeline. It consumes portable RAG or agent traces and emits claim verdicts, minimal evidence spans, citation states, source assessments, abstention decisions, failure labels, primary root causes, and evidence-backed repair and regression suggestions. Figure 1 summarizes the pipeline.

Our contributions are: (1) a typed evidence-chain representation and diagnostic contract; (2) a production-path benchmark protocol with same-ID comparisons, case-bootstrap intervals, eleven registered ablation rows, and fail-closed claim gates; (3) frozen pre-review results on ContextTrace-Diag-150 and a 200-case RAGTruth sample; and (4) a preregistered human actionability protocol plus a clearly separated 1,410-judgment LLM-simulated stress test. Independent human review is pending, so we make dataset-specific pre-review claims only.

2 Related Work

RAG evaluation. RAGAS decomposes automatic evaluation into faithfulness, answer relevance, and context relevance (Es et al., 2024). ARES uses synthetic training data and lightweight judges for context relevance, answer faithfulness, and answer relevance (Saad-Falcon et al., 2024). DeepEval provides a software-

oriented collection of LLM evaluation metrics (Confident AI, 2026). These tools make automatic testing practical, but their outputs are not interchangeable with root-cause, citation, and span diagnostics. Our baseline tables therefore score only common outputs and mark unsupported fields N/A.

Diagnostic benchmarks. RAGChecker introduces fine-grained retriever and generator diagnostics (Ru et al., 2024). RAGTruth supplies word-level hallucination annotations across RAG tasks (Niu et al., 2024), while CRAG evaluates factual and temporally dynamic retrieval settings (Yang et al., 2024). ContextTrace uses RAGTruth as external calibration evidence and CRAG as a secondary track; it adds an explicit serialized chain from claims to repairs. The current external annotations remain review-pending.

Model judges and observability. LLM judges can approximate human preferences (Zheng et al., 2023), but order, self-enhancement, and fairness biases remain material (Wang et al., 2024). We therefore report our model-based review solely as a simulated pilot. At the systems layer, OpenTelemetry defines vendor-neutral trace concepts (Cloud Native Computing Foundation, 2026), and OpenInference extends semantic conventions to AI workloads (Arize AI, 2026). ContextTrace accepts such trace-like records but targets semantic evidence diagnosis rather than telemetry transport.

3 Evidence-Chain Forensics

Let a trace be $T = (q, a, C, Z, M)$, where q is a query, a an answer, $C = \{c_j\}$ retrieved contexts, Z citation edges, and M optional source and execution metadata. Claim decomposition produces $A = \{a_i\}$. For each claim, the verifier emits

$$y_i = (v_i, e_i, z_i, s_i, r_i, f_i), \quad (1)$$

where v_i is a support verdict, e_i a localized evidence span, z_i a citation state, s_i a source-state assessment, r_i a reason, and f_i a claim-level fix. An aggregate diagnosis $D(T)$ contains failure labels, one primary observable root cause, an abstention decision, and a regression target.

Evidence-chain forensics imposes three invariants. *Traceability*: a supported verdict

points to source text. *Coverage honesty*: unavailable outputs remain N/A rather than zero or inferred. *Repair locality*: root causes describe the earliest observable component supported by the trace, not hidden implementation state. For example, missing evidence can support “retrieval miss” only when a supplied corpus or retrieval contract establishes that the evidence should have been available; otherwise the safer label is “should have abstained” or “corpus gap.”

Figure 2 illustrates a contradiction chain. The evidence says refunds must be requested within 30 days while the answer claims 60 days. The chain preserves the contradictory span and links it to answer overreach and a specific answer constraint. This structure is intended to make diagnoses inspectable by both developers and reviewers.

4 System

ContextTrace accepts a JSON trace containing query, answer, contexts, citations, and optional metadata. The implementation normalizes the trace, decomposes the answer into claims, discovers candidate evidence, evaluates support and contradiction, checks citation alignment, assesses source metadata, decides whether abstention was required, and infers a primary root cause. Each stage returns typed JSON consumed by the next stage and by the final report.

Claim verification combines lexical anchors with deterministic semantic normalization for entities, quantities, dates, negation, and common relational paraphrases. Contradiction checks are independent of citation checks: a claim may be contradicted by one source while citing a different source, or it may be supported by the retrieved set while citing the wrong source. Evidence-span localization records the source ID and minimal matched text. Source assessment uses freshness, authority, and canonical-source metadata only when provided.

The diagnosis layer aggregates claim records into observable failure types and a primary root cause. Repair planning ties a suggested change to cited evidence and emits a regression-test target rather than claiming the change has already worked. Production profiles expose ci-

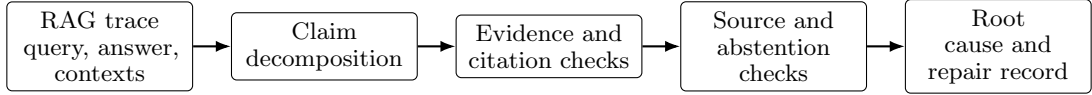


Figure 1: ContextTrace production pipeline. Each stage emits a serializable record used by downstream diagnosis and audit.

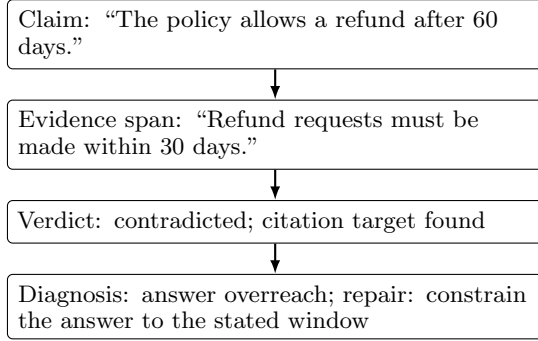


Figure 2: Illustrative evidence-chain trace. The example describes the output schema and is not a scored benchmark result.

tation, contradiction, abstention, source, root-cause, and span modules as explicit switches; ablations therefore execute the shipping verifier rather than a parallel research implementation.

The default pipeline is local, deterministic, and free of API cost. Model-based judges are optional. A provider/model/prompt/cache/cost tuple must be frozen before such outputs enter a result table. No qualifying NLI-only or judge-only configuration existed at freeze time, so both remain registered as unavailable.

5 Benchmark Protocol

We preregister case sets, profiles, seeds, metrics, and claim gates in a machine-readable experiment specification. Prediction, scoring, review, and claim authorization are separate stages (Figure 3). Every bundle records case-ID hashes, candidate hashes, output hashes, dependency versions, runtime, and reported cost.

Datasets. ContextTrace-Diag-150 contains 150 public-document traces covering support, contradiction, abstention, citation, source, and root-cause behavior. The external primary track is a deterministic 200-case sample derived from RAGTruth (Niu et al., 2024). ARES and CRAG adapters provide secondary calibration tracks. The 500-case engineering suite supports

module regression and ablation analysis; it is not independent validation.

Metrics. Primary detection uses failure-label macro-F1 and dangerous false-green rate, where a false green is a trace with a material labeled failure that is reported as clean. Diagnostic outputs use root-cause accuracy, citation-error F1, and token-level evidence-span overlap. Claim-verdict macro-F1 is reported separately because answer-level RAGTruth labels do not always determine an unambiguous claim-level projection. We report deterministic 95% case-bootstrap intervals with 400 draws for the frozen experiment.

Review and claims. The reviewer packets hide predictions, expected labels, original IDs, and private keys. Human sign-off is a distinct gate. Scored, assisted, or LLM-simulated outputs cannot become “independent” through naming. The broad claim gate validates bundle integrity, external coverage, review status, thresholds, confidence intervals, same-ID competitor coverage, and Diag-150 sign-off. A single failed check blocks broad SOTA wording.

6 Experiments

We ask whether ContextTrace detects failures (RQ1), localizes diagnostic structure (RQ2), depends on each pipeline stage (RQ3), and improves repair decisions (RQ4). The full command evaluates the frozen Diag-150 and RAGTruth inputs, scores the same-ID RAGAS prediction file, runs all executable ablations, and regenerates tables and manifests. Quick runs are harness checks and are never paper results.

Table 1 reports the pre-review external results. ContextTrace obtains 0.955 failure-label macro-F1, 0.955 root-cause accuracy, 1.000 citation-error F1, 0.786 span overlap, and zero dangerous false greens on the 200 RAGTruth cases. Diag-150 scores are 1.000 for failure, root-cause, and citation metrics, with 0.950 span overlap. These estimates are calibration

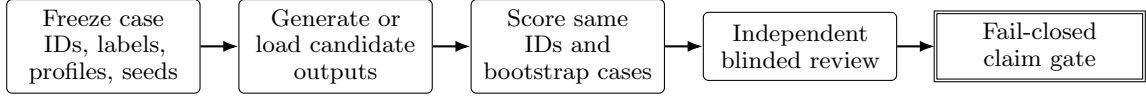


Figure 3: Benchmark protocol. Scoring completion alone does not authorize a broad claim; review status is a separate machine-checked gate.

evidence because independent review is incomplete.

Table 2 compares common outputs on identical RAGTruth IDs. ContextTrace has higher failure-label F1 and lower false-green rate than the cached RAGAS row, while RAGAS has higher claim-verdict F1 (0.533 versus 0.337). RAGAS does not expose matched root-cause, citation, or span fields in this cache; we report N/A rather than attributing implicit zeros. Consequently, the results do not establish broad dominance.

7 Ablations

Table 3 reports eleven registered variants on 500 engineering cases. Nine execute through production profile switches. Lexical- and semantic-only verification each obtain 0.419 failure F1 and a 0.232 false-green rate, showing that claim matching alone is insufficient for trace-level diagnosis. Removing citation alignment lowers failure F1 to 0.831 and raises false greens to 0.122. Removing source trust/freshness produces the largest available full-pipeline degradation: 0.708 failure F1 and 0.112 false greens.

Contradiction checks primarily affect claim decisions: without them, claim-verdict F1 falls to 0.706 and failure F1 to 0.812. Removing abstention logic leaves citation and claim metrics intact but lowers root-cause accuracy to 0.372. Removing the root-cause classifier leaves detection unchanged while correctly making root-cause accuracy unavailable. Removing evidence-span localization makes span overlap unavailable and slightly changes claim and failure behavior because localized evidence also constrains downstream matching.

NLI-only and judge-only rows are N/A. Downloading a model after viewing results would violate the frozen contract; pinning these variants is future work rather than a reason to omit their registered rows.

8 RQ4: Diagnosis Actionability

The preregistered human study contains 40 stratified failure cases and compares a semantic score/claim output against the full evidence-chain report. Option order is counterbalanced and condition names are hidden. At least three non-author reviewers must rate diagnosis correctness, repair actionability, specificity, evidence sufficiency, preference, confidence, and time. No human responses are currently available; the primary human RQ4 outcome is therefore N/A.

We separately execute a controlled LLM-simulated protocol stress test using three role prompts: strict evidence, developer actionability, and skeptical ARR review. Each role evaluates all 40 cases under raw trace (A), score-only (B), and evidence-chain (C) settings, yielding 360 RQ4 judgments. The model is pinned to `gpt-4.1-nano-2025-04-14`; strict schema validation produces zero parse failures. Private labels are loaded only after responses are frozen.

Table 4 shows the simulated results. Setting C raises normalized root-cause accuracy from 0.067/0.117 (A/B) to 0.908 and lowers false greens from 0.625/0.550 to 0.400. However, mean actionability falls from 3.767 (A) and 3.608 (B) to 3.333 (C). Thus the evidence-chain condition does not improve every registered proxy, and the preregistered positive simulated-pilot claim is not supported. The fix-correctness value is an operational proxy, not a human rating. Figure 4 summarizes the protocol.

We include this controlled LLM-simulated review and actionability pilot as a stress test of the review protocol. These results are reported separately and do not replace independent human validation.

9 Error Analysis

The frozen RAGTruth run contains nine primary failure-label misses and no dangerous false greens under the current labels. Table 5

Dataset	Review status	Cases	Failure F1	Root cause	Citation F1	Span overlap	False green
RAGTruth	assisted_review_pending_independent	200	0.955	0.955	1.000	0.786	0.000
ContextTrace-Diag-150	pending_independent	150	1.000	1.000	1.000	0.950	0.000

Table 1: Table1 Main Results. Review status and unavailable measurements are shown explicitly.

System	Review status	Cases	Failure F1	Claim F1	Root cause	Citation F1	Span overlap	False green
ContextTrace	assisted_review_pending_independent	200	0.955	0.337	0.955	1.000	0.786	0.000
RAGAS	same-ID cached baseline	200	0.152	0.533	N/A	N/A	N/A	0.300

Table 2: Table2 Baselines. Review status and unavailable measurements are shown explicitly.

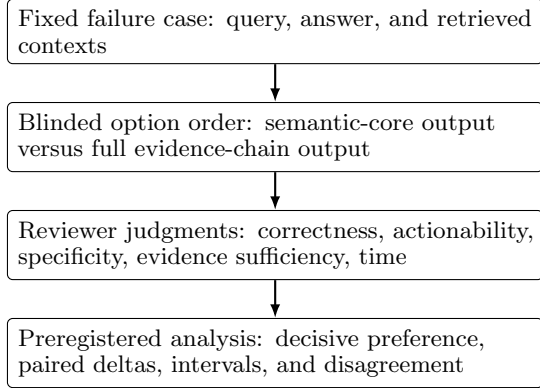


Figure 4: RQ4 study settings. The protocol is frozen; responses have not yet been collected.

separates observed clusters from categories that were not measured. Four clean cases are over-flagged, two contradictions are missed, and nine primary root causes differ. The same-ID score-only baseline produces 60 dangerous false greens. Span breadth, multi-span attribution, numeric/date normalization, and stale-source behavior lack dedicated external tags and remain `not_measured`; judge parse failures are `not_run`.

The simulated annotation stress test sharpens the uncertainty rather than resolving it. Three agents disagree on 142 of 200 RAGTruth cases (Fleiss $\kappa = 0.033$) and 29 of 150 Diag cases ($\kappa = 0.679$). Simulated majorities disagree with 137 and 77 frozen labels respectively. All 214 suggestions remain unapplied. Under a sensitivity-only replacement of those labels, combined classification accuracy falls from 0.974 to 0.380 and macro-F1 from 0.978 to 0.139. This instability demonstrates why model consensus cannot be substituted for independent adjudication.

The high RAGTruth failure score and low claim-verdict score also expose a scope boundary: detecting an answer-level failure does not guarantee a correct claim-by-claim decompo-

sition. Final analysis will preserve raw reviewer responses, report disagreements, and apply only separately versioned, rationalized corrections.

10 Discussion

The experiments support a bounded conclusion: ContextTrace provides a reproducible local evidence-forensics contract and strong pre-review failure-detection behavior on the frozen samples. They do not establish that it is universally more accurate than RAG evaluators, nor that its suggested fixes improve human debugging. The matched RAGAS comparison favors different systems on different common metrics, and unsupported diagnostic outputs are not a fair basis for dominance claims.

The ablations suggest that diagnostic modules matter jointly. Source state and citation alignment substantially reduce false greens, while abstention logic is important for root-cause assignment. This supports exposing the chain rather than only its final score. At the same time, the simulated RQ4 result shows that more structure can reduce perceived actionability: a detailed report may be correct yet cognitively expensive or insufficiently prioritized. Human study design should therefore measure time and preference, not only diagnosis match.

Claim boundary. We claim a benchmarked, local-first evidence-chain forensics system with frozen pre-review results. We do not claim broad SOTA, independent review, human actionability improvement, or real-world truth verification. Dataset-specific metrics must retain sample, baseline, and review status. The machine-readable gate currently blocks broad SOTA while review is pending.

Variant	Availability	Review status	Failure F1	Root cause	Citation F1	Span overlap	False green	p95 ms	Cost / 100
Full ContextTrace	available	engineering regression benchmark	1.000	1.000	1.000	0.862	0.000	38.978	0.000
Lexical-only verifier	available	engineering regression benchmark	0.419	N/A	N/A	N/A	0.232	3.366	0.000
Semantic-only verifier	available	engineering regression benchmark	0.419	N/A	N/A	N/A	0.232	20.541	0.000
No citation module	available	engineering regression benchmark	0.831	0.878	N/A	0.862	0.122	36.000	0.000
No contradiction checks	available	engineering regression benchmark	0.812	0.980	1.000	0.862	0.008	41.366	0.000
No abstention logic	available	engineering regression benchmark	0.875	0.372	1.000	0.862	0.000	38.715	0.000
No root-cause classifier	available	engineering regression benchmark	1.000	N/A	1.000	0.862	0.000	45.393	0.000
No source trust/freshness	available	engineering regression benchmark	0.708	0.888	1.000	0.862	0.112	17.151	0.000
No evidence-span localization	available	engineering regression benchmark	0.982	0.998	1.000	N/A	0.000	57.817	0.000
NLI-only mode	not.available	engineering regression benchmark	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Judge-only mode	not.available	engineering regression benchmark	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Table 3: Table3 Ablations. Review status and unavailable measurements are shown explicitly.

Study	Review status	Cases	Reviewers / agents	Root cause	Fix proxy	Actionability	False green
Preregistered paired actionability study	pending three independent reviewers	40	0/3 humans	N/A	N/A	N/A	N/A
Simulated A: raw trace	LLM-simulated pilot; not human validation	40	3	0.067	0.008	3.767	0.625
Simulated B: score only	LLM-simulated pilot; not human validation	40	3	0.117	0.017	3.608	0.550
Simulated C: evidence chain	LLM-simulated pilot; not human validation	40	3	0.908	0.342	3.333	0.400

Table 4: Table4 Rq4 Actionability. Review status and unavailable measurements are shown explicitly.

Reproducibility. The full workflow pins case IDs, profiles, bootstrap seed, candidate files, dependency versions, command, hashes, runtime, and cost metadata. Table 6 summarizes the run. A post-simulated-review rerun from a committed revision preserves all case hashes and reported metrics, applies zero corrections, and includes sensitivity artifacts separately.

11 Conclusion

ContextTrace represents RAG failures as inspectable chains connecting answer claims, evidence spans, citations, source conditions, abstention, root causes, and repairs. Its deterministic pipeline, production-path ablations, same-ID baseline handling, and fail-closed evidence gates make the evaluation auditable. Frozen pre-review results are promising, but model-review sensitivity and the negative simulated actionability result reinforce the need for independent human validation. The next empirical milestone is completion of the blinded RAGTruth, Diag-150, and three-reviewer RQ4 protocols.

12 Limitations

The primary limitation is evidence status: no independent reviewer has signed off the RAGTruth or Diag-150 labels, and the human RQ4 study has not begun. All headline metrics are therefore pre-review calibration evidence. The LLM-simulated pilots use one model family under three prompts, not independent humans or independent model populations. Their low RAGTruth agreement and poor label-replacement sensitivity preclude us-

ing them as ground truth.

ContextTrace-Diag-150 is synthetic and designed around the implemented taxonomy. The external RAGTruth experiment uses a 200-case deterministic sample rather than the entire corpus. Answer-level labels do not always map cleanly to atomic claim verdicts, which contributes to the 0.337 claim-verdict macro-F1. The evidence-span metric is token overlap and does not directly measure boundary minimality or multi-span recall.

The verifier uses deterministic lexical and semantic heuristics rather than a general entailment model. It may fail on implicit reasoning, long dependency chains, domain-specific numerical relations, multilingual traces, and evidence distributed across many documents. Source trust and freshness require metadata that many production traces omit. Grounding against supplied contexts does not establish external factual truth.

NLI-only and judge-only ablations are unavailable because no qualifying frozen model artifacts were registered. Latency measurements are workstation-specific. The same-ID baseline cache does not expose every ContextTrace diagnostic and its original generation cost is unavailable. Finally, the anonymous artifact can reproduce software results but cannot redistribute human identities, private condition keys, or future licensed response data.

13 Ethics Statement

ContextTrace can process traces locally, reducing the need to transmit prompts, answers, or retrieved documents to an external evaluator. Local execution does not remove obligations

Cluster	Review status	Count	Example	What happened / likely cause	Planned fix	Affects broad claim
Supported claim overflagged	assisted_review_pending_independent	4	ragtruth_9604	Support threshold or label boundary	Independent adjudication and calibrated claim decomposition	True
Contradiction missed	assisted_review_pending_independent	2	ragtruth_121	Taxonomy boundary or implicit contradiction	Adjudicate labels; add contradiction regression cases	True
Evidence span too broad	assisted_review_pending_independent	not_measured	N/A	No dedicated breadth annotation	Add span-boundary error tags during review	unknown
Multi-span evidence failure	assisted_review_pending_independent	not_measured	N/A	Distributed evidence is not separately tagged	Add multi-span annotations and recall metric	unknown
Root cause confused	assisted_review_pending_independent	9	ragtruth_121	Failure label and upstream cause are difficult to separate	Adjudicate root causes and report agreement	True
Numeric/date normalization error	assisted_review_pending_independent	not_measured	N/A	No dedicated numeric/date error tag	Add typed normalization error labels	unknown
Citation mismatch ambiguity	assisted_review_pending_independent	not_measured	N/A	No citation mismatch was observed in the primary errors	Retain citation-specific regression coverage	False
State/source-trust limitation	assisted_review_pending_independent	not_measured	N/A	RAGTruth does not independently label source freshness	Evaluate on a temporally labeled dataset	unknown
Score-only baseline false-green	same-ID cached baseline	60	aggregate	Score-only output lacks diagnostic coverage	Report same-ID false-green rate and N/A fields	True
Judge parse failure	not_run	not_run	N/A	Judge-only mode is unavailable without a frozen provider	Pin provider/model/cache/cost before running	unknown

Table 5: Table5 Error Analysis. Review status and unavailable measurements are shown explicitly.

Item	Review status	Value	Status
Source revision	pre_review_paper_candidate	15cf09f9f9d7bffd6285dae2603edad12d8a3ff9	frozen
Command	pre_review_paper_candidate	python benchmarks/contexttrace_bench/reproduce_arr_tables.py --full --output-dir 'benchmarks/contexttrace_bench/out/arr_full'	full
Python	pre_review_paper_candidate	3.8.10	recorded
Runtime seconds	pre_review_paper_candidate	249.847	recorded
Bootstrap	pre_review_paper_candidate	400 / 20260612	frozen
Context Trace API cost	pre_review_paper_candidate	0.0	local
Candidate cost	same-ID cached baseline	N/A	cached/unreported
Paper result eligible	pending independent review	False	pending review

Table 6: Table6 Reproducibility. Review status and unavailable measurements are shown explicitly.

around access control, retention, dataset licenses, or personal information contained in traces. Users should minimize stored content, restrict report access, and avoid publishing sensitive evidence spans.

Automatic diagnoses can be wrong and should not be the sole basis for decisions in medical, legal, financial, employment, or other high-impact settings. A “supported” result means supported by supplied evidence, not true in the world. Repair suggestions require engineering validation and a recaptured passing trace.

The simulated-review pilot is explicitly labeled to prevent model outputs from being misrepresented as human judgments. Future human participation must be voluntary, appropriately reviewed where required, compensated consistently, and reported without identifying reviewers in the anonymous submission. Raw responses and disagreements should be retained rather than selectively removed.

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